

Naveena Wimalarathne

Associate AI/ML Engineer
naveenawimalarathne01@gmail.com | +94 71 011 6305 | LinkedIn | GitHub | My Portfolio



Profile

Computer Science graduate with hands-on experience in Artificial Intelligence, Machine Learning, Big Data, Data Science, and full-stack software development. Completed a BSc (Hons) in Computer Science at the University of Sri Jayewardenepura, with a strong academic foundation in Computer Science, Statistics, and Mathematics. Experienced as a Big Data and Data Science Intern at Mobitel, with practical exposure to Hadoop, PySpark, RAG-based chatbots, LLM-powered enterprise systems, and AI application deployment. Passionate about building intelligent, scalable, and business-ready AI solutions for real-world industry problems.

Technical Skills

- Python, Java, C, R
- Machine Learning, NLP, Computer Vision, RAG, LLMs, LangChain, YOLO, Scikit-learn, TensorFlow, OpenCV, Pandas, NumPy, Matplotlib, RAGAS, DeepEval
- Hadoop, HDFS, MapReduce, PySpark, Tableau, SSRS
- HTML, CSS, JavaScript, Bootstrap, React, Figma, MySQL, SQLAlchemy, Thymeleaf, Spring Boot, Flask, FastAPI, Neo4j, MongoDB
- Jupyter Notebook, Google Colab, VS Code, IntelliJ, GNU Octave, MATLAB, Minitab
- Git, GitHub, GitLab, Jenkins, Docker, Postman, XAMPP, SonarQube, Selenium

Education

BSc (Hons) in Computer Science University of Sri Jayewardenepura	2026
Current GPA: CS - 3.51 Statistics - 3.3 Overall - 3.4	
Higher National Diploma in IT University of Colombo School of Computing	2024
GCE Advanced Level Examination Visakha Vidyalaya, Colombo 5	2020
Combined Maths - A Chemistry - B Physics - C	
Z Score - 1.36	

Work Experience

Mobitel (Pvt) Ltd Intern - Big Data & Data Science	Dec 2025 – Jun 2026
– Worked with Big Data and Data Science team on telecom data analytics, data processing, and AI-related enterprise use cases. – Gained hands-on exposure to Hadoop, HDFS, MapReduce, PySpark, and large-scale telecom data processing workflows. – Trained a YOLO-based NIC verification system handling crop detection, rotation correction, and image similarity matching. – Built an AI-Assisted Legal Management System automating legal case handling and document workflows — awarded 1st Runner-Up at the SLT AI Assisted Coding Challenge. – Architected an Agentic AI server monitoring pipeline using Kafka, LogAI, Logstash, Fluentd, and OpenSearch for real-time anomaly detection. – Deployed a Docker-based AI CV screening system integrating OCR extraction and LLM-based candidate ranking. – Developed LLM/RAG chatbot components including document chunking strategies, vector retrieval, and response generation pipelines. – Delivered BI dashboards and data analysis reports using Python, SQL, SSRS and Tableau to support business decision-making.	
Dchem Learning Centre Teacher Assistant (IT & Administration)	Nov 2021 – Jan 2023
DFCC Bank PLC Trainee Banking Assistant	Mar 2021 – Jun 2021

Projects

LLM based Intelligent Assistant for Navigating Procurement Guidelines in Sri Lanka Final Year Research
Developed an LLM-based chatbot using a Retrieval-Augmented Generation (RAG) architecture, with document chunking and vector database integration, to assist users in navigating Sri Lanka’s government procurement guidelines.
AI Assisted Legal Management System (Mobitel (Pvt) Ltd) Python LLMs RAG Semantic Search SQLite LangChain
Contributed to an AI-assisted legal workflow management system designed to automate legal case handling, document validation, approval workflows, and legal template generation and awarded 1st Runner-Up in the AI Assisted Coding Challenge at Sri Lanka Telecom.

NIC Verification System (Mobitel (Pvt) Ltd)
Python | YOLO | AWS Rekognition

Built an automated identity document pipeline with YOLO-based detection, rotation correction, crop alignment, and image similarity matching for Sri Lankan NIC validation.

Server Performance Monitoring Using Agentic AI (Mobitel (Pvt) Ltd)
Agentic AI | Kafka | LogAI | OpenSearch | Logstash | Fluentd | Fluent Bit | Time-Series Monitoring

Designed an Agentic AI log analytics architecture for real-time anomaly detection and automated operational insights from server logs.

NEXU - News Verification Robot using Crowdsourcing
Python | Reddit API | PRAW | Groq | Raspberry Pi | Arduino

Developed a real-time news verification system as part of the Human-Robotics Interaction Project, which scrapes Reddit data and analyzes content credibility using crowdsourced signals and sentiment analysis. Integrated Raspberry Pi and Arduino to enable physical interaction and movement based on the system’s verification outcomes.

Stroke Risk Prediction – EDA & ML Pipeline
Python | Scikit-learn | XGBoost | SMOTE | Pandas | Matplotlib

Built a full analytics pipeline (descriptive, diagnostic, predictive, prescriptive) on the Kaggle Stroke Prediction dataset. Addressed class imbalance using SMOTE, compared Logistic Regression, SVM, Random Forest, and XGBoost (best accuracy: 90%), and applied Differential Evolution for prescriptive risk minimization.

Medicine Recommendation System
Python | Jupyter Notebook | HTML | Pandas | Scikit-learn | Flask

Developed a symptom-based medicine recommendation engine with a responsive frontend using Flask. The system suggests appropriate medications based on user-input symptoms.

Hand Gesture Game for Kids
Python | OpenCV

Developed an interactive game controlled by real-time hand gestures using Python and OpenCV. The system tracks finger movements to enable touchless gameplay, enhancing user interaction and demonstrating gesture-based controls in gaming.

Glow Hatty - Smart Lamp
Arduino IDE | Edge Impulse

Designed and built a smart lamp that uses machine learning to detect objects (books vs. phones) and automatically adjusts lighting based on user activity.

University Hall Booking System
Java | Spring Boot | Thymeleaf | HTML | CSS | MySQL | Docker | FreeSQL

Developed a full-stack web application for hall reservations at the University of Sri Jayewardenepura. Features include user login, real-time hall availability, booking submission, admin approval panel, and a user dashboard. Deployed using Docker with a cloud-hosted MySQL database on FreeSQL.

Referees

Snr. Prof. R G N Meegama
Senior Professor
Department of Computer Science
University of Sri Jayewardenepura
+94 71 838 8055
rgn@sjp.ac.lk

Prof. Ravindra De Silva
Professor
Department of Computer Science
University of Sri Jayewardenepura
+94 77 746 4000
ravi@sjp.ac.lk

Extra Curricular Activities

- General Secretary of the Computer Science Association 2025/2026
- Assistant Secretary of the Computer Science Association 2024/2025
- Co-Chief Organizer of the Algorhythm 2024 - Department Night
- First Year Committee Member of the Society of Computer Science 2022/2023
- Member of the Rotaract Club of University of Sri Jayewardenepura 2022/2023
- Committee Member of the Leo Club of Kottawa Central Golden City 2021/2022
- Committee Member of the Drug Prevention Club of Visakha Vidyalaya 2019/2020
- Junior Prefect (2016/2017)
- Team Member of the School Table Tennis Team (Under 15/17)